Physics-Informed LSTM for Snowmelt-Driven Streamflow Forecasting to Support Hydropower Grid Integration

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***Abstract*—Accurate streamflow forecasting is essential for hydropower scheduling and grid integration, yet snowmelt-dominated basins exhibit abrupt regime transitions—including melt onset and rain-on-snow events—that cause conventional prediction intervals to lose coverage precisely when operators need reliable bounds most. Although physics-informed and data-driven models have improved point predictions, existing uncertainty quantification methods assume stationary or exchangeable residuals, a condition that snowmelt hydrology systematically violates. This paper presents a Physics-Informed Long Short-Term Memory (PI-LSTM) model with Dual Adaptive Conformal Prediction (DACP) for daily streamflow forecasting and calibrated uncertainty quantification across five snowmelt-dominated watersheds in the Western United States. The PI-LSTM integrates a site-adaptive water balance constraint into the training loss, improving R^2 over a standard LSTM baseline at all five sites with up to 40% RMSE reduction. DACP dynamically adapts interval width to regime transitions, achieving 93.8–94.6% empirical coverage at the 95% target across all five watersheds—even at the most challenging site where point predictions explain only 27.6% of variance. These bounds enable grid operators to schedule reserves against calibrated lower bounds rather than worst-case assumptions.**

***Index Terms*—Hydropower forecasting, long short-term memory, physics-informed machine learning, uncertainty quantification, snowmelt runoff**

I. Introduction

Hydropower provides approximately 6% of total electricity generation in the United States and remains the largest source of renewable energy in several Western states [1]. Unlike wind and solar resources, hydropower generation is primarily governed by seasonal snowpack accumulation and melt processes, creating a unique forecasting challenge: operators must predict streamflow weeks to months ahead to optimize reservoir operations, schedule generation, and coordinate with other resources on the grid [2]. In snowmelt-dominated basins, the relationship between meteorological inputs and discharge is highly nonlinear. Rapid temperature increases can trigger abrupt melt onset, rain-on-snow events produce sharp discharge spikes, and glacial melt contributes irregular baseflow patterns [3]. These dynamics create substantial uncertainty in hydropower generation forecasts, directly affecting day-ahead scheduling accuracy and grid reliability. Moghaddasi et al. showed that winter snow water equivalent is a leading indicator of spring and summer streamflow in snow-dominated western U.S. basins, with the strongest SWE–streamflow relationship at lookbacks of 60–90 days [4]. Traditional hydrological models such as the Hydrologiska Byrans Vattenbalansavdelning (HBV) model and degree-day approaches require extensive calibration and domain expertise, and often struggle to capture the rapid transitions characteristic of spring snowmelt [5]. Data-driven approaches using machine learning have gained traction: recurrent neural networks, particularly Long Short-Term Memory (LSTM) networks [6], have demonstrated strong performance on streamflow prediction tasks [7], [8]. However, purely data-driven models may learn physically implausible relationships, especially when training data are limited [9]. Physics-informed machine learning (PIML) addresses this limitation by embedding physical constraints into the learning process [10]. Maharjan et al. compared a physics-informed LSTM against a standard LSTM and a process-based model (SWAT) for snowpack and streamflow simulation in a snow-fed Californian catchment, and found that the physics-informed model provided the most robust and balanced performance, particularly in minimizing bias [11]. Read et al. demonstrated the effectiveness of process-guided deep learning for environmental prediction tasks [12]. Despite these advances, several gaps remain in the literature. First, most LSTM-based streamflow studies produce only point predictions without uncertainty quantification, which is a critical limitation for risk-aware grid operations. Moreover, climate-driven non-stationarity introduces distribution shifts that violate the exchangeability assumptions of traditional conformal methods. These shifts cause prediction intervals to lose coverage precisely during the abrupt melt transitions and discharge spikes when reliable bounds are most needed. Second, few studies evaluate physics-informed approaches across multiple watersheds with diverse flow regimes simultaneously. Third, the connection between improved streamflow forecast accuracy and operational grid value (e.g., dispatch scheduling, curtailment reduction) is rarely quantified. This

paper addresses these gaps with the following contributions:

- A PI-LSTM architecture that integrates a water balance constraint into the training loss using site-adaptive physics weighting, evaluated across five snowmelt-dominated watersheds spanning diverse hydrological regimes.
- To address the distribution shifts and abrupt regime transitions that cause traditional uncertainty quantification to fail in snowmelt basins, we employ Dual Adaptive Conformal Prediction (DACP) [13], which provides distribution-free, calibrated prediction intervals at multiple coverage targets (70%, 80%, 90%, 95%).
- Demonstration that DACP generalizes across five watersheds with distinct flow regimes—achieving $\geq$93.8% coverage at the 95% target without site-specific tuning—even when point predictions are poor ($R^2 = 0.276$), decoupling uncertainty reliability from model accuracy.
- Comparison against a standard LSTM baseline (identical architecture, $\lambda = 0$) showing consistent R^2 improvement at all five sites with up to 40% RMSE reduction.

II. Related Work

Snowmelt Modeling and Streamflow Prediction. Classical snowmelt models include the degree-day method and energy balance approaches [14]. While physically grounded, these models require calibrated parameters for each basin and struggle with rapid melt transitions. Kiewiet et al. modeled spatially distributed surface water input in a rain–snow transition catchment and found that annual discharge was correlated with the magnitude of precipitation and spring snow and rain rather than with snowfall fraction, underscoring the importance of precipitation magnitude and timing for streamflow generation [3]. This paper leverages precipitation-derived features (7-day cumulative precipitation and antecedent precipitation index) informed by these findings. *Machine Learning for Hydropower Forecasting.* LSTM networks [6] have emerged as a leading approach for hydrological time series prediction. Kratzert et al. demonstrated that a single LSTM model can match or exceed the performance of calibrated conceptual models across hundreds of basins [7]. Feng et al. extended this work with attention mechanisms to improve peak flow prediction [8]. While Transformer-based architectures have shown promise in some time series tasks, LSTMs remain the dominant architecture for rainfall-runoff modeling due to their proven ability to learn long-term hydrological memory from sequential meteorological forcing [7]. However, these studies focus on streamflow without connecting results to hydropower generation scheduling. This paper bridges the hydrology-power systems gap by framing streamflow prediction as a hydropower generation forecasting problem with uncertainty bounds suitable for grid dispatch. *Uncertainty Quantification for Power System Forecasting.* Prediction intervals are essential for risk-aware power system operations. Conformal prediction has gained attention as a distribution-free framework for constructing prediction intervals [15]. Gibbs and Candès [23] introduced Adaptive Conformal Inference (ACI), which uses a single learning-rate regulator to maintain coverage under distribution shift; however, a single regulator cannot independently control both the calibration window partition and the quantile level, limiting responsiveness to the abrupt regime transitions characteristic of snowmelt hydrology. The DACP framework, proposed in [13], extends ACI with two adaptive regulators, a data partitioning regulator and a quantile regulator, to handle non-exchangeable data and distribution shifts. To the best of our knowledge, this is the first application of DACP to snowmelt-driven streamflow forecasting with multi-watershed evaluation, demonstrating that the framework generalizes effectively to diverse basins with distinct flow characteristics.

TABLE I
Study Site Characteristics

Site	Peak Flow (cfs)	Regime
Bunchgrass Meadow	∼4,500	High snowmelt
Easy Pass	∼275	Alpine, GHI-driven
Paradise	∼35	Rain-on-snow
Touchet	∼680	Seasonal melt
Swift Creek	∼30	Glacial melt

III. Methodology

A. Study Area and Data

Five snowmelt-dominated watersheds in the Western United States were selected to represent diverse hydrological regimes (Table I). Regime classifications are based on our analysis of site hydrographs and associated metadata. All datasets contain 1,461 daily observations from 2019 to 2022, sourced from the USDA Natural Resources Conservation Service SNOTEL network [20] and United States Geological Survey (USGS) streamflow gauges via the Water Data for the Nation portal [21]. Surface meteorological observations (daily maximum and minimum temperature and precipitation) were obtained from the NOAA GHCN-Daily archive [22], and Global Horizontal Irradiance (GHI) from the National Solar Radiation Database (NSRDB) [24]. Discharge data are recorded at daily resolution in cubic feet per second (cfs).

B. Snowmelt Modeling

The relationship between temperature and snowmelt is captured through a simplified degree-day formulation [14] embedded in the physics loss term. The daily snowmelt rate M_t (mm/day) is estimated as:

$$M_t = D_f \cdot \max(T_t - T_b,\ 0) \tag{1}$$

where D_f is the degree-day factor (mm/°C/day), T_t is the mean daily air temperature, and T_b is the base temperature threshold. This estimate, combined with precipitation and evapotranspiration, forms the water balance constraint in the physics-informed loss function.

C. Hydropower Generation Model

For run-of-river hydropower installations typical of the study watersheds, generation is directly proportional to streamflow:

$$P_t = \eta \cdot \rho \cdot g \cdot Q_t \cdot h \tag{2}$$

where P_t is power output (W), η is turbine efficiency, ρ is water density (kg/m^3), g is gravitational acceleration (m/s^2), Q_t is volumetric flow rate (m^3/s), and h is the effective hydraulic head (m). Since η, ρ, g, and h are approximately constant for a given facility, accurate streamflow forecasting directly determines generation forecast accuracy.

D. PI-LSTM Forecasting Architecture

The PI-LSTM processes eight input features over a 14-day sliding window. Four features are raw meteorological variables (maximum temperature, minimum temperature, precipitation, GHI), and four are engineered: 7-day cumulative precipitation (prcp_7d), antecedent precipitation index (API), and sinusoidal day-of-year encodings ($\sin(2\pi d/365)$, $\cos(2\pi d/365)$). Two site-specific feature sets were identified through Optuna hyperparameter optimization [16]: F1 uses prcp_7d (Easy Pass, Paradise, Touchet) and F2 uses API (Bunchgrass Meadow, Swift Creek). The architecture consists of two stacked LSTM layers [6] with 128 hidden units each, followed by a 4-head multi-head attention mechanism and fully connected layers (128→64→32) with ReLU activations and 0.2 dropout. Training minimizes a combined loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda \cdot \mathcal{L}_{\text{physics}} \tag{3}$$

where $\mathcal{L}_{\text{data}}$ is a site-specific data loss (Huber for Easy Pass, Paradise, Touchet; Nash–Sutcliffe Efficiency [19] for Bunchgrass Meadow; mean squared error for Swift Creek) and $\mathcal{L}_{\text{physics}}$ enforces the daily water balance [9]:

$$\mathcal{L}_{\text{physics}} = \frac{1}{N}\sum_{t=1}^{N}\left(Q_t - [P_t + M_t - ET_t - \Delta S_t]\right)^2 \tag{4}$$

where Q_t is discharge, P_t is precipitation, M_t is snowmelt, ET_t is evapotranspiration estimated via a simplified temperature–precipitation approximation, and ΔS_t is the daily change in snow water equivalent (WESD). The physics penalty λ is optimized per site via Optuna hyperparameter search [16] on validation R^2, searching $[0.001, 0.5]$ on a log scale. For Bunchgrass Meadow, $\lambda = 0$ because the $\log(1+Q)$ transform used to stabilize training on its 0–4,500 cfs range causes the water balance gradient to collapse to near-zero in log space, adding noise rather than useful signal. For Swift Creek, $\lambda = 0$ because at consistently low flows (5–30 cfs) the physics loss magnitude dominates the data loss and degrades prediction accuracy.

E. Training, Validation, and Evaluation

A walk-forward prediction strategy is employed: the model is initially trained on early data and periodically retrained as the training window expands, eliminating temporal data leakage. Key training parameters include patience of 30 epochs, maximum 300 epochs per retraining pass, and a fixed random seed for reproducibility. Models are evaluated using three metrics:

$$\text{NSE} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{5}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2} \tag{6}$$

$$\text{MAE} = \frac{1}{N}\sum|y_i - \hat{y}_i| \tag{7}$$

Traditional uncertainty quantification methods for time series often assume stationarity or exchangeability of residuals, but snowmelt-driven streamflow violates these assumptions: climate-driven distribution shifts, abrupt melt transitions, and sudden discharge spikes cause conventional prediction intervals to undercover during the most critical periods. The DACP framework [13] addresses this by constructing adaptive, distributionally robust prediction intervals using a 180-day rolling calibration window with two adaptive regulators: a partition regulator t_m that controls the calibration/reference split, and a quantile regulator α_m that adjusts the interval width. Both update at each time step based on whether the previous observation was covered, enabling the intervals to widen rapidly after misses and contract gradually during stable periods. The DACP framework generalizes effectively across all five sites without requiring site-specific parameter tuning. The baseline is a *Standard LSTM* using the identical architecture (2-layer LSTM, 128 hidden units, 4-head attention, same feature sets) but trained with data loss only ($\lambda_{\text{phys}} = 0$), directly isolating the value of the physics constraint.

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS FIVE SITES

Site	Model	R^2	RMSE (cfs)	MAE (cfs)
Bunchgrass M.	Std. LSTM	0.735	399.2	276.3
	PI-LSTM	**0.790**	**238.7**	**174.1**
Easy Pass	Std. LSTM	0.131	19.6	14.7
	PI-LSTM	**0.545**	**15.7**	**6.5**
Paradise	Std. LSTM	0.651	3.35	1.54
	PI-LSTM	**0.677**	**4.31**	**2.09**
Touchet	Std. LSTM	0.644	15.7	5.8
	PI-LSTM	**0.893**	**11.0**	**4.3**
Swift Creek	Std. LSTM	0.207	4.45	1.56
	PI-LSTM	**0.276**	**5.93**	**1.37**

IV. RESULTS AND DISCUSSION

A. Point Prediction Performance

Table II presents the PI-LSTM performance compared to the Standard LSTM baseline across all five sites. The PI-LSTM improves RMSE over the Standard LSTM at three of five sites: Bunchgrass Meadow (238.7 vs. 399.2 cfs, 40.2% reduction), Touchet (11.0 vs. 15.7 cfs, 29.9%), and Easy Pass (15.7 vs. 19.6 cfs, 19.8%). At Swift Creek, the PI-LSTM achieves higher R^2 (0.276 vs. 0.207) but elevated RMSE

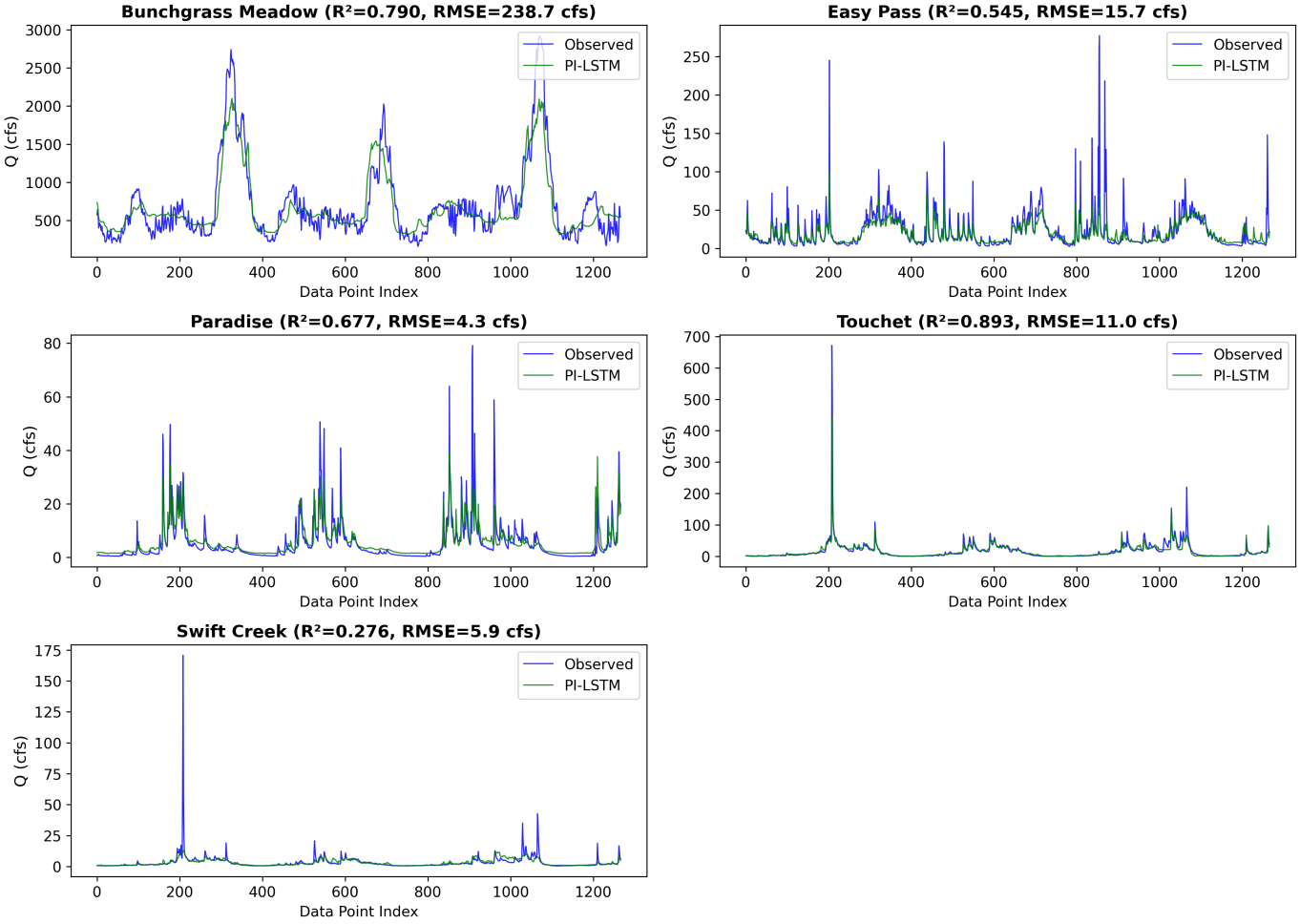


Fig. 1. Observed (blue) versus PI-LSTM predicted (green) daily streamflow for all five sites across the 2019–2022 test period. Each panel shows R^2 and RMSE. The PI-LSTM captures seasonal snowmelt peaks across diverse flow regimes, from high-discharge Bunchgrass Meadow (∼4,500 cfs) to low-flow Swift Creek (∼30 cfs).

(5.93 vs. 4.45 cfs), reflecting the narrow flow range where the physics constraint improves variance explanation at the cost of a small absolute error increase. The PI-LSTM achieves higher R^2 at all five sites, confirming that the water balance constraint provides meaningful regularization. Fig. 1 shows the observed versus PI-LSTM predicted streamflow for all five sites across the 2019–2022 test period. At Paradise, the PI-LSTM achieves higher R^2 (0.677 vs. 0.651) but higher RMSE (4.31 vs. 3.35 cfs) than the Standard LSTM. This is attributable to Paradise's rain-on-snow flash regime: ***the physics constraint encourages the model to attempt capturing brief, sharp discharge spikes rather than smoothing them***, improving variance explanation (R^2) at the cost of larger absolute errors during low-flow periods when even small mispredictions inflate RMSE. Touchet achieves the highest accuracy ($R^2 = 0.893$) due to its clean seasonal cycle with a single dominant snowmelt peak per year. The Huber loss function was critical here: the site features a massive peak event (∼680 cfs) that dwarfs normal discharge levels of 20–50 cfs, and Huber loss prevents this extreme event from dominating training gradients. Swift Creek is the most challenging ($R^2 = 0.276$), as its glacial melt regime produces irregular, multi-peaked hydrographs that the LSTM struggles to predict. The low R^2 reflects the narrow flow range (5–30 cfs) where even small absolute errors explain little variance. Easy Pass ($R^2 = 0.545$) presents a distinct challenge: the site transitions from near-zero winter flow (snow-locked) to sharp melt-driven spikes in late spring, driven by rapid solar radiation (GHI) increases. The PI-LSTM systematically underestimates during May and June because the 14-day input window does not fully capture the abrupt solar transition that triggers melt onset. Despite this, the 19.8% RMSE improvement over the Standard LSTM confirms that the physics constraint helps anchor predictions during these rapid transitions.

B. Uncertainty Quantification

Table III presents DACP coverage rates and average interval widths across four target levels. All five sites achieve empirical coverage within 1–2 percentage points of each nominal target across all coverage levels, demonstrating well-

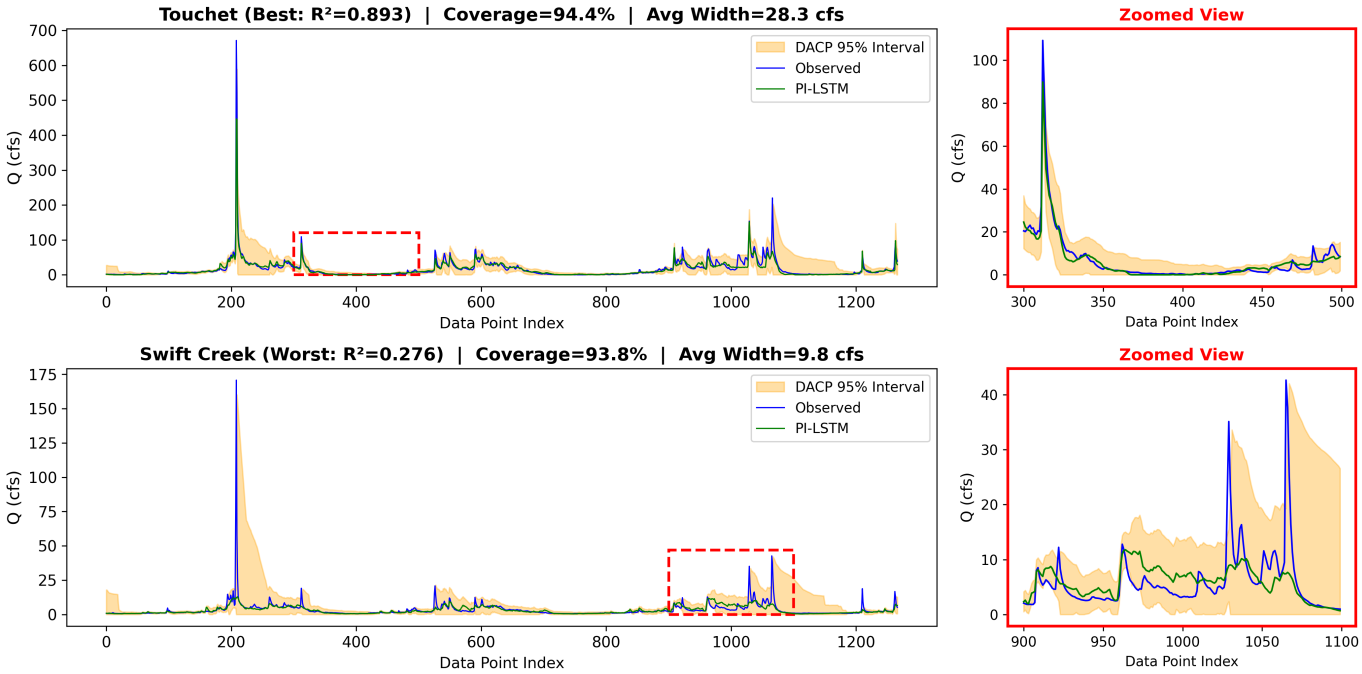


Fig. 2. PI-LSTM predictions with 95% DACP intervals for Touchet (best $R^2 = 0.893$, top) and Swift Creek (worst $R^2 = 0.276$, bottom). Left: full time series. Right: zoomed spring snowmelt period (red dashed box). Intervals widen around discharge peaks and contract during stable baseflow. Despite Swift Creek's low R^2, DACP achieves 93.8% coverage via wider intervals; Touchet enables tighter bounds (28.3 vs. 9.78 cfs).

TABLE III
DACP COVERAGE (%) AND INTERVAL WIDTH (CFS) ACROSS TARGETS

Site	70%	80%	90%	95%
Bunchgrass	70.6 / 490	80.7 / 597	90.3 / 743	94.6 / 816
Easy Pass	69.7 / 12.3	79.8 / 16.9	89.7 / 29.1	94.5 / 47.0
Paradise	70.0 / 4.71	79.2 / 6.36	89.7 / 10.1	94.5 / 14.9
Touchet	71.3 / 9.42	79.9 / 13.1	89.8 / 19.1	94.4 / 28.3
Swift Creek	70.5 / 2.36	79.9 / 3.46	89.1 / 5.96	93.8 / 9.78

180-day rolling window; coverage / avg width format.

calibrated uncertainty bounds. A key implementation detail is the update speed ratio of the DACP regulators: the miss-reaction speed $\mu_2 = 0.50$ is set approximately 17× larger than the coverage-reaction speed $\mu_1 = 0.03$, ensuring the system widens intervals aggressively after misses but tightens them only gradually. Initial tuning with near-symmetric speeds achieved only 80–85% coverage at the 95% target. The most notable result is the decoupling between point prediction accuracy and uncertainty quantification reliability. Swift Creek achieves only $R^2 = 0.276$ but DACP still provides 93.8% coverage at the 95% target by appropriately widening intervals. Conversely, Touchet ($R^2 = 0.893$) achieves the tightest relative interval width (28.3 cfs, ∼4.2% of peak discharge), demonstrating that better predictions enable tighter but still reliable intervals. Fig. 2 illustrates this contrast. ***Unlike fixed-width intervals, DACP dynamically widens bounds during discharge spikes and contracts them during stable baseflow, producing operationally useful intervals that are neither wastefully wide nor dangerously narrow.*** Across all sites and targets, empirical coverage falls within 1–2 percentage points of the nominal level, from high-discharge Bunchgrass Meadow (95% width = 816 cfs) to low-flow Swift Creek (95% width = 9.78 cfs), without site-specific recalibration.

C. Seasonal Performance Analysis

Performance varies by season, most prominently at sites with large seasonal peaks: at Bunchgrass Meadow and Touchet, spring RMSE (March–June) increases by 38% and 21% over the annual average, respectively, while summer baseflow (July–September) yields substantially lower errors. The DACP intervals adapt accordingly, widening during melt onset and contracting during stable periods—critical for grid operators facing the highest uncertainty during spring runoff when hydropower capacity is greatest.

D. Limitations

The physics constraint could not be applied at Bunchgrass Meadow and Swift Creek ($\lambda = 0$) due to log-transform gradient collapse and scale mismatch, respectively, and the ET proxy is a simplified approximation. The run-of-river generation model assumes constant head. The 14-day input window may miss slow snowpack accumulation [4], the four-year record limits climate non-stationarity assessment, and all results use a fixed random seed.

E. Implications for Grid Operations

In hydropower-dependent regions of the Western U.S. [17], [18], the calibrated DACP prediction intervals enable probabilistic dispatch: operators can schedule thermal reserves based

on the lower bound of the 95% interval rather than worst-case assumptions. At Touchet, the PI-LSTM's 30% RMSE reduction combined with calibrated uncertainty bounds can support more efficient day-ahead scheduling. The fact that DACP achieves reliable coverage ($\geq$93.8%) even at Swift Creek ($R^2 = 0.276$) confirms that operators can trust the uncertainty bounds regardless of point prediction accuracy.

V. Conclusion

This paper presented a Physics-Informed LSTM model with Dual Adaptive Conformal Prediction for streamflow forecasting across five snowmelt-dominated watersheds. The PI-LSTM achieves up to 40% RMSE reduction over the Standard LSTM baseline, with R^2 improvements at all five sites (0.276–0.893), confirming the value of the water balance physics constraint. This constraint improves prediction quality for sites with sufficient flow dynamic range, while site-specific data loss functions (Huber, NSE, MSE) prevent extreme events from dominating training. Critically, DACP addresses the key operational bottleneck: climate-driven distribution shifts render traditional uncertainty quantification unreliable during volatile periods when trustworthy bounds matter most. DACP's adaptive regulators maintain 93.8–94.6% coverage at the 95% target across all five sites without site-specific recalibration, and the decoupling between point prediction accuracy and coverage reliability—most evident at Swift Creek (93.8% coverage despite $R^2 = 0.276$)—confirms operationally trustworthy bounds regardless of base model performance. Future work includes extending input sequences to 30–60 days to capture slow snowpack accumulation, coupling with reservoir operations models for end-to-end generation forecasts, and exploring spatial-temporal correlations between sites to further reduce prediction interval width.

Acknowledgment

The authors gratefully acknowledge the support of the RaDA Lab and the Texas Artificial Intelligence Research Institute at the University of Texas at Dallas.

References

[1] U.S. Energy Information Administration, "Electricity explained: Electricity in the United States," 2024. [Online]. Available: https://www.eia.gov/energyexplained/electricity/electricity-in-the-us.php

[2] U.S. Bureau of Reclamation, "Hydropower program: Managing water in the West," U.S. Dept. Interior, Washington, DC, USA, Tech. Rep., 2023.

[3] L. Kiewiet, E. Trujillo, A. Hedrick, S. Havens, K. Hale, M. Seyfried, S. Kampf, and S. E. Godsey, "Effects of spatial and temporal variability in surface water inputs on streamflow generation and cessation in the rain–snow transition zone," *Hydrol. Earth Syst. Sci.*, vol. 26, no. 11, pp. 2779–2796, 2022, doi: 10.5194/hess-26-2779-2022.

[4] P. Moghaddasi, K. Gavahi, and H. Moradkhani, "Snow drought to hydrologic drought progression using machine learning and probabilistic analysis," *Sci. Rep.*, vol. 15, no. 1, art. no. 20918, 2025, doi: 10.1038/s41598-025-05978-y.

[5] S. Bergström, "The HBV model," in *Computer Models of Watershed Hydrology*, V. P. Singh, Ed. Highlands Ranch, CO, USA: Water Resources Publications, 1995, pp. 443–476.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997, doi: 10.1162/neco.1997.9.8.1735.

[7] F. Kratzert, D. Klotz, C. Brenner, K. Schulz, and M. Herrnegger, "Rainfall–runoff modelling using long short-term memory (LSTM) networks," *Hydrol. Earth Syst. Sci.*, vol. 22, no. 11, pp. 6005–6022, 2018, doi: 10.5194/hess-22-6005-2018.

[8] D. Feng, K. Fang, and C. Shen, "Enhancing streamflow forecast and extracting insights using long-short term memory networks with data integration at continental scales," *Water Resour. Res.*, vol. 56, no. 9, art. no. e2019WR026793, 2020, doi: 10.1029/2019WR026793.

[9] A. Karpatne, G. Atluri, J. H. Faghmous, M. Steinbach, A. Banerjee, A. Ganguly, S. Shekhar, N. Samatova, and V. Kumar, "Theory-guided data science: A new paradigm for scientific discovery from data," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 10, pp. 2318–2331, 2017, doi: 10.1109/TKDE.2017.2720168.

[10] J. D. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Comput. Surv.*, vol. 55, no. 4, pp. 1–37, 2022, doi: 10.1145/3514228.

[11] S. Maharjan, W. Li, R. Thomas, S. Fazli, A. Ansari, H. Morgan, A. Elgendy, M. Allali, and H. El-Askary, "Physics-informed deep learning reveals climate-driven snowpack decline and threatens ecological water availability in a Californian snow-fed catchment," *Ecol. Inform.*, vol. 92, art. no. 103526, 2025, doi: 10.1016/j.ecoinf.2025.103526.

[12] J. S. Read, X. Jia, J. Willard, A. P. Appling, J. A. Zwart, S. K. Oliver, A. Karpatne, G. J. A. Hansen, P. C. Hanson, W. Watkins, M. Steinbach, and V. Kumar, "Process-guided deep learning predictions of lake water temperature," *Water Resour. Res.*, vol. 55, no. 11, pp. 9173–9190, 2019, doi: 10.1029/2019WR024922.

[13] Y. Zong, S. Badakhshan, J. Zhang, and Y. Xu, "Trustworthy uncertainty quantification via distributionally-robust dual adaptive conformal prediction," *J. Mech. Des.*, vol. 148, no. 2, art. no. 021703, 2025, doi: 10.1115/1.4069683.

[14] R. Hock, "Temperature index melt modelling in mountain areas," *J. Hydrol.*, vol. 282, no. 1–4, pp. 104–115, 2003, doi: 10.1016/S0022-1694(03)00257-9.

[15] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *J. Mach. Learn. Res.*, vol. 9, pp. 371–421, 2008.

[16] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2019, pp. 2623–2631, doi: 10.1145/3292500.3330701.

[17] J. D. Hunt, E. Byers, Y. Wada, S. Parkinson, D. E. H. J. Gernaat, S. Langan, D. P. van Vuuren, and K. Riahi, "Global resource potential of seasonal pumped hydropower storage for energy and water storage," *Nat. Commun.*, vol. 11, no. 1, art. no. 947, 2020, doi: 10.1038/s41467-020-14555-y.

[18] National Renewable Energy Laboratory, "Hydropower's contribution to grid flexibility," NREL, Golden, CO, USA, Tech. Rep. NREL/TP-5700-80174, 2021.

[19] J. E. Nash and J. V. Sutcliffe, "River flow forecasting through conceptual models part I—a discussion of principles," *J. Hydrol.*, vol. 10, no. 3, pp. 282–290, 1970, doi: 10.1016/0022-1694(70)90255-6.

[20] U.S. Dept. of Agriculture, Natural Resources Conservation Service, "SNOwpack TELemetry (SNOTEL) Network," [Online]. Available: https://www.wcc.nrcs.usda.gov/snow/

[21] U.S. Geological Survey, "USGS Water Data for the Nation: National Water Information System," [Online]. Available: https://waterdata.usgs.gov/nwis

[22] NOAA National Climatic Data Center, "Global Historical Climatology Network - Daily (GHCN-Daily), Version 3," NOAA Nat. Climatic Data Center, Asheville, NC, USA. [Online]. Available: https://doi.org/10.7289/V5D21VHZ

[23] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 1660–1672.

[24] National Renewable Energy Laboratory, "National Solar Radiation Database (NSRDB)," [Online]. Available: https://nsrdb.nrel.gov